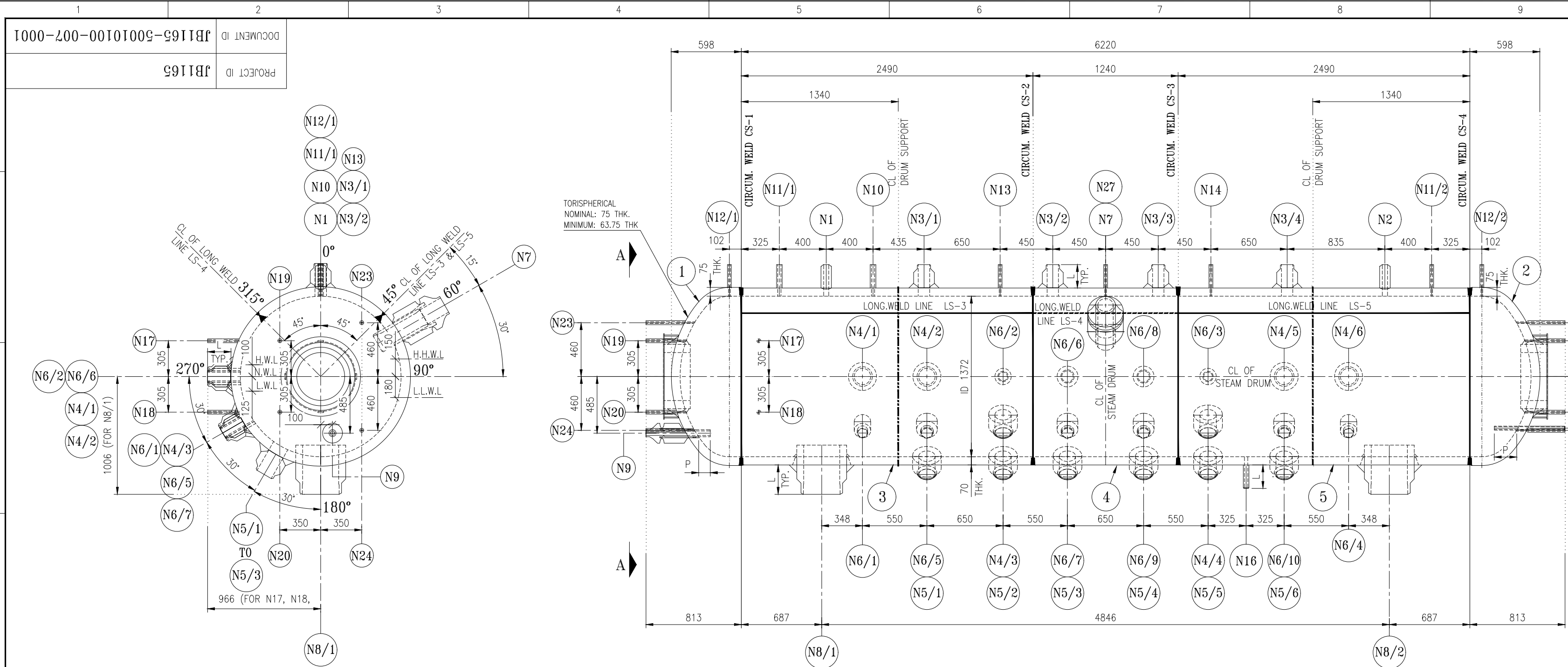
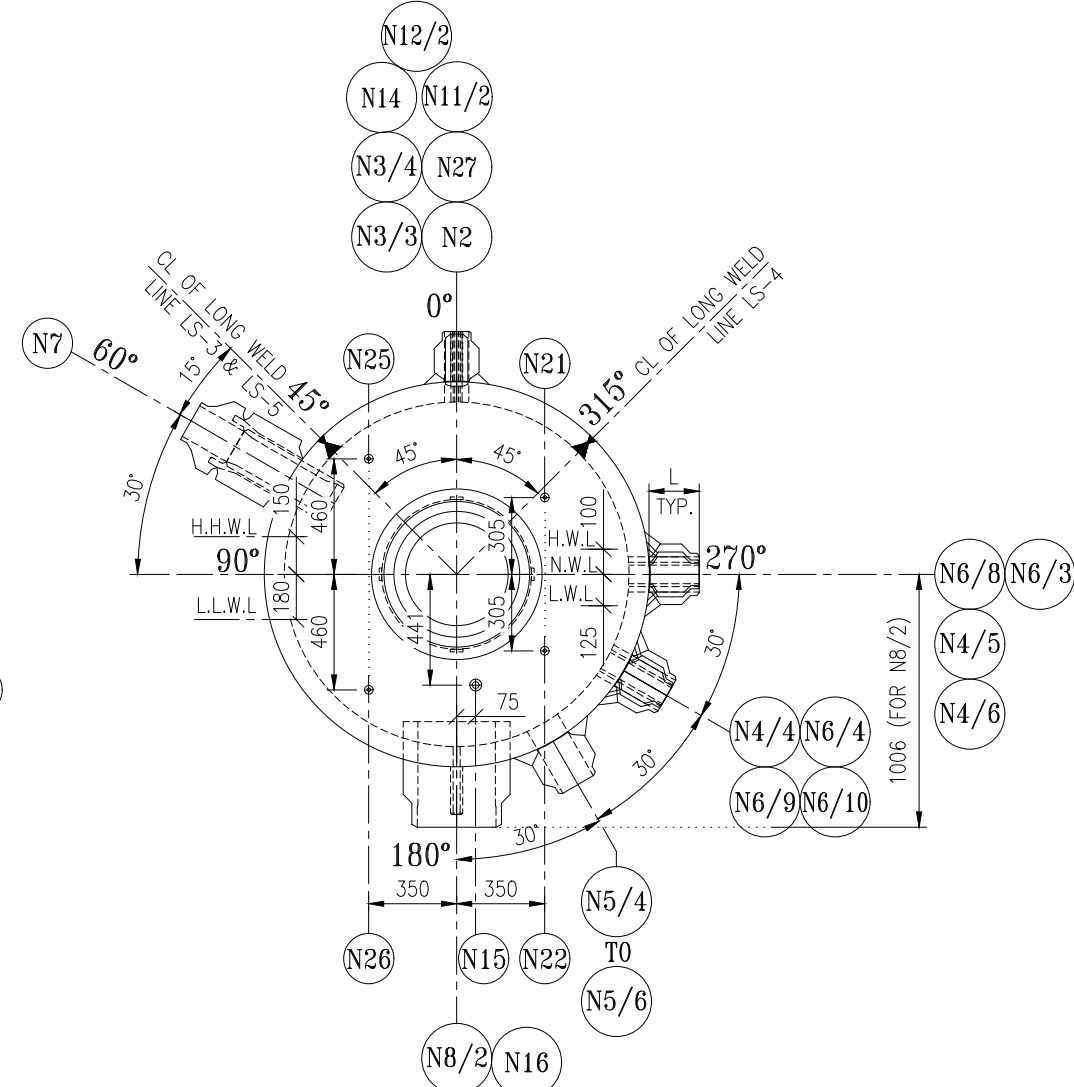


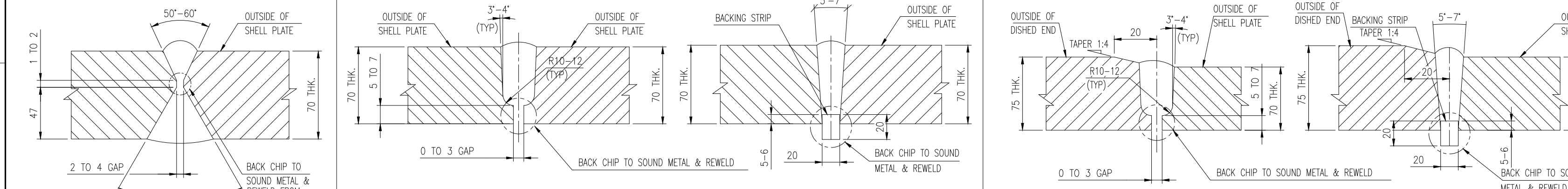


ELEVATION
ARRANGEMENT OF STEAM DRUM [VIEW FROM FRONT OF BOILER]



VIEW - B B
[VIEW FROM RHS OF BOILER]

VIEW - A A
[VIEW FROM LHS OF BOILER]



WELD PREPARATION OF LONGITUDINAL SEAM
FOR STEAM DRUM
(SEAM MK. Nos. LS-3 TO LS-5)

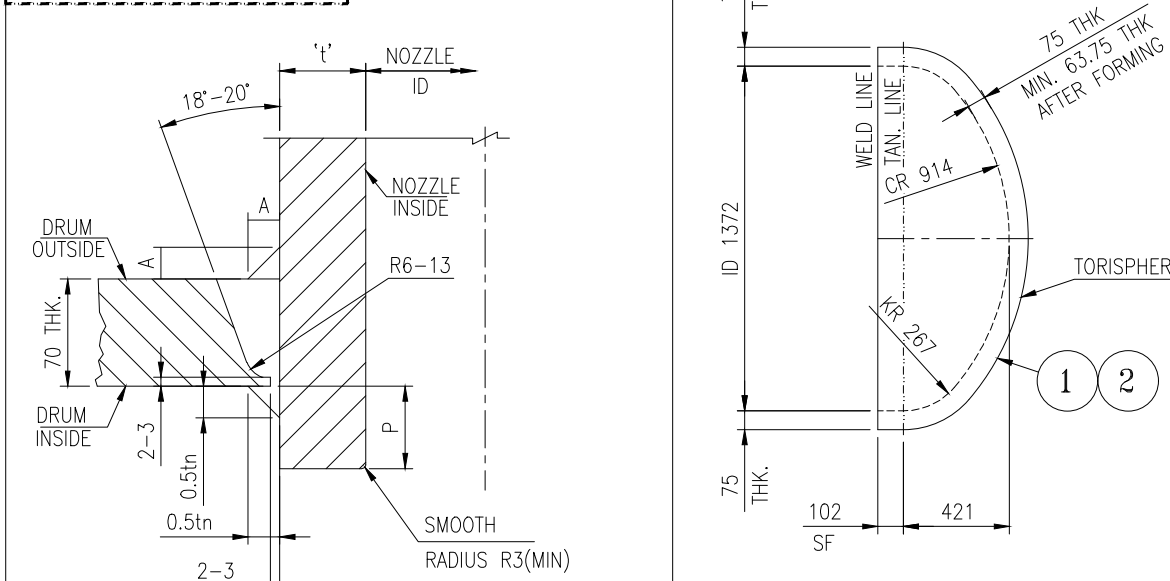
WELD PREPARATION OF CIRCUMFERENTIAL SEAM
(SEAM MK. Nos. CS-2 & CS-3)

ALTERNATE DETAIL

WELD PREPARATION OF CLOSING SEAM
FOR STEAM DRUM MARK Nos. CS-1 & CS-4

ALTERNATE DETAIL

IBR FIG : XII/85E J1
IBR CHAPTER - XII

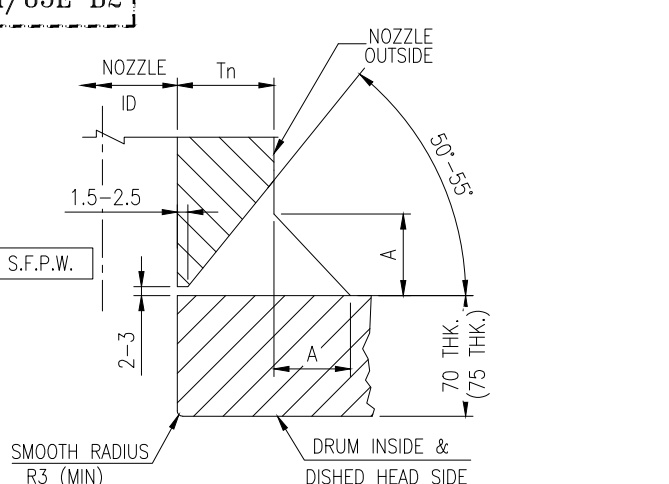


WELDING DETAIL OF NOZZLE TO DRUM
(FOR MK.No. N8/1 & N8/2)

STEAM DRUM
DISHED END DETAIL

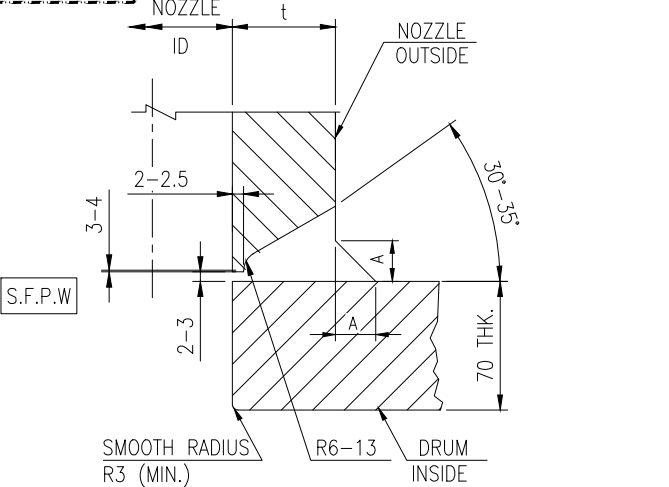
TABLE - 1										
S.Nos.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (mm)	Nozzle Thickness (t) (mm)	Connecting Pipe Thickness (mm)	Projection from End of Drum (mm)	Projection from End of Drum (mm)	Projection from End of Drum (mm)
1	N1	SAFETY VALVE NOZZLE - 1	STEAM DRUM SHELL	1	114.3	34.93	-	200	-	250
2	N2	SAFETY VALVE NOZZLE - 2	STEAM DRUM SHELL	1	114.3	34.93	-	200	-	250
3	N3/1 TO N3/4	SATURATED STEAM STUB	STEAM DRUM SHELL	4	178	40.41	8.56	200	-	250
4	N4/1 TO N4/6	COMBUSTOR SIDE WALL RISER STUB	STEAM DRUM SHELL	6	240	50.12	14.27	200	-	250
5	N5/1 TO N5/6	COMBUSTOR FRONT WALL RISER STUB	STEAM DRUM SHELL	6	240	50.12	14.27	200	-	250
6	N6/1 TO N6/4	SEPARATOR SIDE WALL RISER STUB	STEAM DRUM SHELL	4	140	33.17	7.62	200	-	250
7	N6/5 TO N6/10	SEPARATOR FRONT & REAR WALL RISER STUB	STEAM DRUM SHELL	6	178	42.98	11.13	200	-	250
8	N7	FEED INLET NOZZLE WITH THERMAL SLEEVE	STEAM DRUM SHELL	1	305	60.00	REF. DRG. 009-0001	-	-	REF. DRG. No.009-0001
9	N8/1 & N8/2	DOWN COMER NOZZLE	STEAM DRUM SHELL	2	426	59.03	23.83	250	-	470
10	N9	CHEMICAL INJECTION NOZZLE WITH THERMAL SLEEVE	STEAM DRUM DISHED END	1	170	46.4	REF. DRG. 009-0001	-	100	485
11	N10	AIR VENT NOZZLE	STEAM DRUM SHELL	1	48.3	10.15	5.08	200	-	250
12	N11/1 & N11/2	PR. TRANSMITTER NOZZLE	STEAM DRUM SHELL	2	33.4	9.09	4.55	200	-	250
13	N12/1	PR. GAUGE NOZZLE (LOCAL)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	200	-	250
14	N12/2	REMOTE PR. GAUGE AT FLOOR	STEAM DRUM DISHED END	1	33.4	9.09	4.55	200	-	250
15	N13	INST. TEST GAUGE NOZZLE	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	-	250
16	N14	NITROGEN BLANKING	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	-	250
17	N15	CBD NOZZLE	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	75	441
18	N16	IBD NOZZLE	STEAM DRUM SHELL	1	48.3	10.15	5.08	200	-	250
19	N17	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	305	332
20	N18	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	305	332
21	N19	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	305
22	N20	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	305
23	N21	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	305
24	N22	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	305
25	N23	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	460
26	N24	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	460
27	N25	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	460
28	N26	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	460
29	N27	SPARE STUB	STEAM DRUM SHELL	1	33.4	9.09	-	200	-	250

IBR CHAPTER : XII
IBR FIG : XII/85E B2



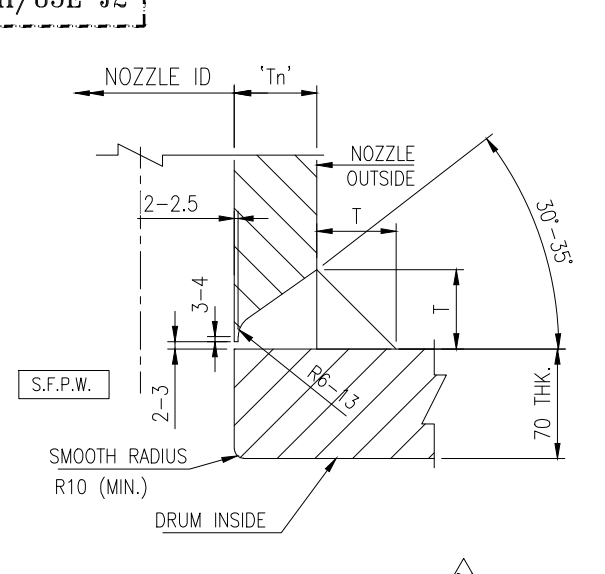
WELDING DETAIL OF NOZZLE TO DRUM (SET ON NOZZLE)
[FOR MARK Nos. N10, N11/1, N11/2, N12/1, N12/2, N13, N14, N16, N17, N18, N19, N20, N21, N22, N23, N24, N25, N26 & N27]
(BRACKETED DIMENSION IS APPLICABLE FOR DISHED END)

IBR CHAPTER : XII
IBR FIG : XII/85E J2



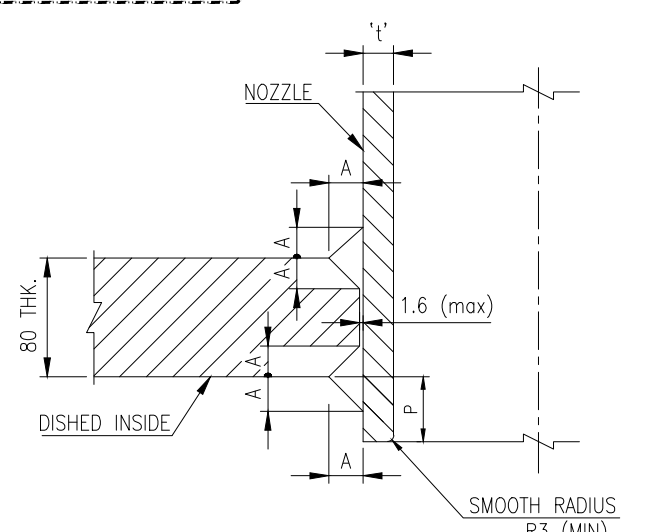
WELDING DETAIL OF NOZZLE TO DRUM (SET ON NOZZLE)
[FOR MARK.Nos. N3/1 TO N3/4, N4/1 TO N4/6, N5/1 TO N5/6 & N6/1 TO N6/10]

IBR CHAPTER : XII
IBR FIG : XII/85E J2

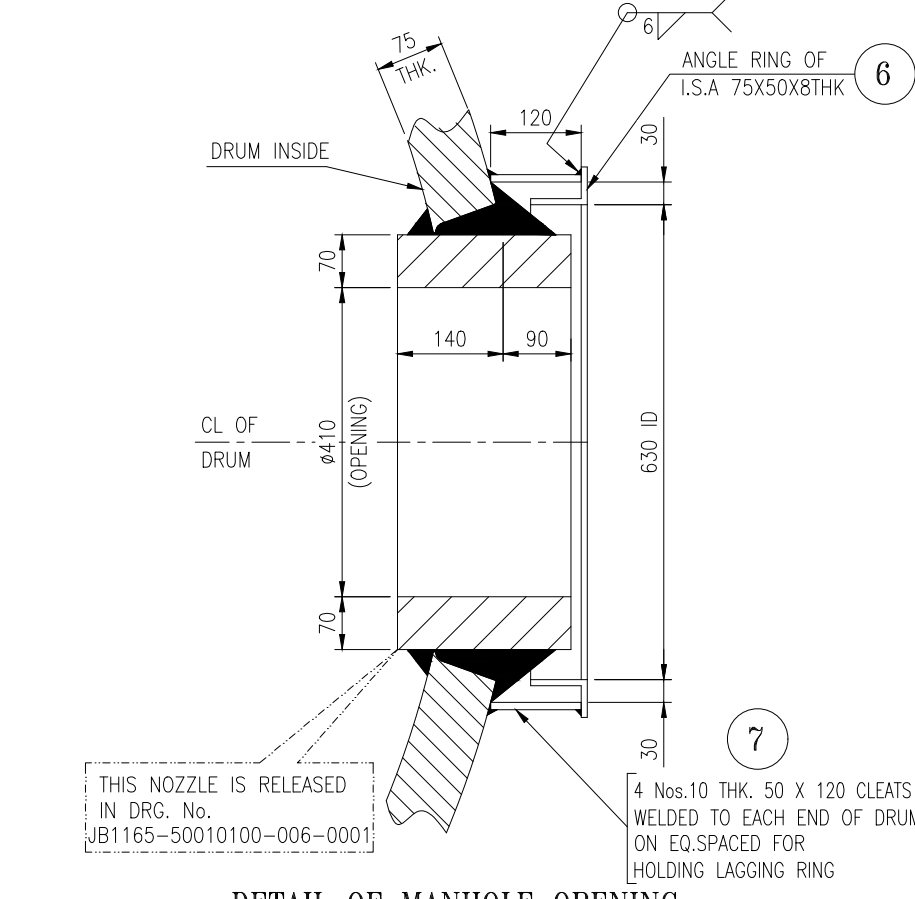


WELDING DETAIL OF NOZZLE TO DRUM (SET ON NOZZLE)
[FOR MARK Nos. N1 & N2]

IBR CHAPTER : VIII
IBR FIG : VIII 365/13



WELDING DETAIL OF NOZZLE TO DRUM
[FOR MK.No. N15]



DETAIL OF MANHOLE OPENING

23. DRUM TO BE STAMPED AS FOLLOWS :-	
ITEM CODE-SDM001- STEAM DRUM	
ATY PROJECTS INDIA LTD., MATHURA (U.P.)	
(A/C ISGEC HEAVY ENGINEERING LTD.)	
WORKS NO - /INT NO. 6666	
YEAR OF MAKE -	
TESTED TO -169.5 Kg/Sq. cm (g)	
TESTED ON -	
MAXIMUM WORKING PRESSURE = 113.0 Kg/Sq. cm (g)	
COMPETENT PERSON'S / INSPECTING AUTHORITY'S OFFICIAL STAMP	
FINAL DATE OF EXAMINATION -	

PAINING NOTES:-

01. FOR PAINTING NOTES REFER SPECIFICATION NO. JB1165-50010100-PSH-0001.

PWHT CYCLE :- (FOR STEAM DRUM)	
A) LOADING TIME (MAX.)	= 3000
B) RATE OF HEATING ABOVE 300°C	= 75°C/hr.
C) SOAKING TEMP.	= 610°C/10°C
D) SOAKING TIME (MIN)	= 2 Hours 20 minutes.
E) RATE OF COOLING UP TO 300°C	= 75°C/hr.
F) UNLOADING TEMP	= 300°C
G) COOLING AFTER UNLOADING	= COOLING IN STILL AIR

DESIGN DATA

CODE OF CONSTRUCTION	IBR 1950 & ITS LATEST AMENDMENTS UP TO - OCTOBER 2019.
MAXIMUM WORKING PRESSURE	113.00 Kg/Sq. cm (g)
MAXIMUM WORKING TEMPERATURE	320 °C
WORKING METAL TEMPERATURE	320 °C
HYDRAULIC TEST PRESSURE	169.500 Kg/Sq. cm (g) (AT SHOP & AS PER IBR AT SITE)
RADIOGRAPHY OF LONG. & CIRCUM. SEAM	100%
STRESS RELIEVING	(REFER HEAT TREATMENT CYCLE TABLE)
EVAPORATION OF BOILER (M.C.R.)	75 TPH
TOTAL HEATING SURFACE AREA OF BOILER (EXCLUDING FORCED FLOW ELEMENT)	1720.00 sq.m (APPROX.)
HEATING SURFACE AREA OF FORCE FLOW SECTION (NON STEAMING)	990.00 sq.m (APPROX.)
HEATING SURFACE AREA OF DRUM	NIL (OUT SIDE GAS PATH)
STEAM PRESSURE AT SUPER HEATER OUTLET	90.00 Kg/Sq. cm (g)
SUPER HEATER OUTLET TEMPERATURE	520±45°C
WEIGHT OF DRUM (EMPTY)	22014.9 Kgs. (APPROX.)
INSPECTION	DIRECTOR OF BOILERS, UTTAR PRADESH/Lloyds (SEE NOTE : 02)

MATERIAL SPECIFICATIONS :

DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg/ Sq.mm)	MIN.CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	60.5	FOR STEAM DRUM
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	57.9	FOR STEAM DRUM
MAN HOLE DOOR	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	77.0	FOR STEAM DRUM

NOTES :-

- THE ZERO DATE OF THIS CONTRACT IS 29.10.2019. HENCE IBR AMENDMENTS UP TO OCTOBER 2019 HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
- INSPECTION BY DIRECTOR OF BOILERS, UTTAR PRADESH/Lloyds CBQ Authorisation, form XIV/A See Regulation 4C(2). BECAUSE MANUFACTURING OF THIS EQUIPMENT SHALL BE DONE AT ATY PROJECT INDIA LIMITED, MATHURA, UTTAR PRADESH.
- ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE SPECIFIED.
- ALL FILLET WELDS ARE 6mm UNLESS OTHERWISE SPECIFIED.
- FOR DIMENSIONAL TOLERANCE REQUIREMENTS PLEASE REFER DOC. NO. GEN-50990100-DSM-002 AND ITS ASSOCIATED DRAWINGS.
- CIRCUMFERENTIAL DIMENSION (ON OUTSIDE TRAMMELS ON DRUM) STEAM DRUM - 4750.1 mm
- LEGEND :
H.H.W.L. : DENOTES HIGH HIGH WATER LEVEL
H.L.W.L. : DENOTES HIGH WATER LEVEL
N.W.L. : DENOTES NORMAL WATER LEVEL
L.W.L. : DENOTES LOW WATER LEVEL
S.F. : DENOTES STRAIGHT FACE
S.F.P.W. : SET ON FULL PENETRATION WELD
T.L. : TANGENTIAL LINE
CS-1ACS-4 : DENOTES CLOSING SEAM
CS-2ACS-3 : DENOTES CIRCUMFERENTIAL SEAM
KR : DENOTES KNUCKLE RADIUS.
LS : LONGITUDINAL SEAM
TL : TANGENTIAL LINE
CR : DENOTES CROWN RADIUS.
KR : DENOTES KNUCKLE RADIUS.

II) SHOP NOTES :

- ALL CIRCUMFERENTIAL DIMENSIONS ARE MEASURED OUTSIDE OF DRUMS
- MANHOLE CIRCULAR TYPE SIZE 4410 FOR DRUMS REF. DRG. No. JB1165-50010100-006-0001
- A MAX. 4th mm FILLET WELD TO BE USED FOR WELDING OF CLEAT BRACKETS AS PER DRG.No. : 011-0001 TO 014-0001 AFTER STRESS RELIVING.
- THIS DRAWING SHALL BE READ IN CONJUNCTION WITH THE FOLLOWING DRGS.
i. FOR NOZZLE & STUDS DETAILS REFER DRG.No. : JB1165-50010100-009-0001
ii. FOR EXTERNAL AND INTERNAL BAFFLES & PIPE REFER DRG.No. : 010-0001 TO 015-0001.
- ALL WELDING TO BE DONE BY IBR QUALIFIED WELDER USING IBR WELDING PROCEDURES
- FOR BOILER DRUMS HAVING SHELL THICKNESS 100mm AND ABOVE FOLLOWING ADDITIONAL TESTING SHALL BE CARRIED OUT:
i. ULTRASONIC EXAMINATION OF ALL BUTT WELD BEFORE AND AFTER PWHT.
ii. MAGNETIC PARTICLE EXAMINATION OF NOZZLES/STUDS TO SHELL WELDING.
- ALL THE PADS WELDED ALL-ROUND ON PRESSURE PART SHALL BE PROVIDED WITH 2 Nos 1/8" NPT(1) TELL-TALE HOLES AT 180° APART ON THE PAD TO RELEASE ENTRAPPED AIR/GAS.
- TOLERANCE FOR FORMED HEADS:-
THE INNER SURFACE OF A TORISPHERICAL SHALL NOT DEVIATE OUTSIDE OF THE SPECIFIED SHAPE BY MORE THAN 1/8" OF NOMINAL INSIDE DIAMETER & INSIDE THE SPECIFIED SHAPE BY MORE THAN 3/8" OF NOMINAL INSIDE DIAMETER
- SUITABLE STIFFENER ARRANGEMENT SHALL BE PROVIDED AT SHOP FOR NOZZLES AT DISHED END TO AVOID DAMAGE DURING TRANSPORTATION.
- MANHOLE DOORS ARE TO BE PLACED INSIDE THE DRUM BEFORE ASSEMBLING DRUM ENDS FOR FINAL CIRCUMFERENTIAL WELD AS PER ORIENTATION DETAIL
- 2mm THK. SS PLATE SHALL BE PROTECTED USING WOODEN BLANKS DURING TRANSPORTATION.
- CRUISE TO HYDRAULIC TESTING FOLLOWING SHALL BE CARRIED OUT
a) ALL WELD SPATTER, WELD STUDS, SCALE, DIRT ETC SHALL BE REMOVED FROM DRUM.
- EXAMINATION SHALL BE THOROUGHLY DRIED/CLEANED AFTER HYDRO TESTING.
- HYDRO TESTING TO BE CARRIED OUT WITH GASKET THAT IS THE SAME AS THE SPECIFIED SERVICE GASKET.
- HYDROTEST WATER SHALL NOT CONTAIN MORE THAN 10ppm SUSPENDED SOLIDS.
- ALL INSIDE EDGES OF NOZZLES AND CONNECTIONS SHALL BE GROUND SMOOTH TO 3 MM RADIIUS.
- STAY RING SHALL BE WELDED IF ANY STAY RING NOTED SHALL BE DRESSED, WELDED & UT CHECKED.
- FOLLOW APPROVED AND APPLICABLE WPS FOR SELECTION OF WELDING CONSUMABLES.
- WELD SURFACES SHALL BE FREE FROM CRACKS, UNDERCUTS, UNNECESSARY GRINDING, ABRUPT RIDGES AND VALLEYS.
- VISUAL INSPECTION OF WELD SHALL BE CARRIED OUT AND NOTED BEFORE OFFERING FOR ANY NON DESTRUCTIVE EXAMINATION.
- HYDRO TEST WATER QUALITY SHALL BE MAINTAINED WITHIN LIMITS SPECIFIED IN DOCUMENT NO.- BLS-PPA-HYD-01-S1.

-- FLATS & BRACKETS ON DRUM --	--	1	70.38	JB1165-50010100-015-0001
-- FLATS & STUDS INSIDE DRUM --	--	1	211.17	JB1165-50010100-012-0001
-- DETAIL OF DRUM NOZZLE --	--	1	203.38	JB1165-50010100-009-0001
-- A&D OF MAN HOLE DOOR --	--	2	SET	JB1165-50010100-006-0001
7 CLEAT FOR RING	IS 2062 Gr. A	PL.10 THK-50 X 120	8	3.77
6 LAGGING RING	IS 2062 Gr. A	ISA 75X50X8 THK (AS PER STD.)	2	33.94
5 SHELL	SA 516 Gr.70	PL.70 THK-2500 X 4820	1	6062.8
4 SHELL	SA 516 Gr.70	PL.70 THK-1250 X 4970	1	3032.27
3 SHELL	SA 516 Gr.70	PL.70 THK-2500 X 4820	1	6062.8
2 DISHED END	SA 516 Gr.70	PL.75 THK-2060 X 2060	1	1821.34
1 DISHED END	SA 516 Gr.70	PL.75 THK-2060 X 2060	1	1819.20

SDM001	STEAM DRUM WITH ATTACHMENTS			P	B215	1	22014.91		
	P.NO.	PART DESCRIPTION	SPECIFICATION	SIZE			P.QTY	P.WT(kg)	REMARK
ITEM CODE	ITEM DESCRIPTION			L.TYPE	L.GROUP	L.QTY	L.WT (Kg) (TOTAL)	REMARK	

CONSULTANT	--								
CLIENT	M/s. SHREAS INDUSTRIES LIMITED.								
DO NOT SCALE IF IN DOUBT PLEASE ASK	THIS DESIGN IS THE PROPERTY OF M/S. ISGEC HEAVY ENGINEERING LTD. AND IS ONLY ALLOWED TO BE USED BY EXPRESSED PERMISSION AND LICENSE FROM M/S. ISGEC HEAVY ENGINEERING LTD.								
FOR APPROVAL	NO	FOR INFORMATION	NO	FOR PRODUCTION	YES	FOR ERECTION	YES		
RESPONSIBLE DEPT.	MECHANICAL	DRAWN BY	V.KAVASKAR	CHECKED BY	R.GAMBAR/R.KUMARAN	APPROVED BY	R.ARUNKUMAR	DATE	19.12.2019
TOTAL WEIGHT IN kg.	22014.91	PROJECT ID	JB1165	ELEMENT ID	50010100	I.W.T. NO.	6666		
YEAR	2019	SHEET SIZE	A1	SCALE	1 : 30				

I.W.T. No. : 6666	
CONSULTANT	
CLIENT	
DO NOT SCALE IF IN DOUBT PLEASE ASK	
FOR APPROVAL	
RESPONSIBLE DEPT.	
MECHANICAL	
DRAWN BY	
V.KAVASKAR	
CHECKED BY	
R.GAMBAR/R.KUMARAN	
APPROVED BY	
R.ARUNKUMAR	
DATE	
19.12.2019	
TOTAL WEIGHT IN kg.	
22014.91	
PROJECT ID	
JB1165	
ELEMENT ID	
50010100	
I.W.T. NO.	
6666	
YEAR	
2019	
SHEET SIZE	
A1	
SCALE	
1 : 30	
SERVICE	
SINGLE DRUM, WATER TUBE, TOP SUPPORTED, CFBC BOILER	
TITLE	
ARRANGEMENT & DETAIL OF STEAM DRUM	
DRAWING DOCUMENT ID	
JB1165-50010100-007-0001	
REVISION	
01	

REVN.	ZONE	DESCRIPTION	REV'D. BY	CHK'D. BY	APPD. BY	DATE	REVN.	ZONE	DESCRIPTION	REV'D. BY	CHK'D. BY	APPD. BY	DATE
1							1						
2							2						
3							3						
4							4						
5							5						
6							6						
7							7						
8							8						
9							9						
10							10						
11							11						
12							12						
13							13						
14							14						
15							15						